Air Flow Control Damper VCD



Description
Aluminum rectangular air flow control damper with opposable blades.

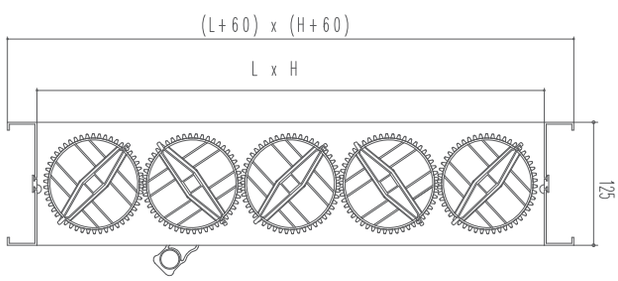
Application
Air flow adjustment in ventilation systems.

Characteristics
The standard control damper is provided with a drive lever and blocking system, an electric drive (actuator) can be delivered on request.
Actuator will be selected according to the size of the damper, voltage and other technical details mentioned in the project (return spring, 0-10V control, etc.)
Control dampers can be made, on request, without drive system - in this case it will be considered the diameter of the blade's drive shaft for coupling the desired actuator. (eg. Control damper without drive lever, prepared for actuator)
Drive lever / actuator operates the opening / closing of the blades through 12 X 12mm and 150mm length shaft .
Aerodynamic blades are a main characteristic of the control damper.
The control damper is provided with gasket seals on both the blades and the frame.
Blade is 100mm height and cannot be adjusted, having the following two cases:
* Case 1: 200 x 200 mm - the control damper consists of two adjustable blades and the final measurements will become 200 x 212 mm [W (H + 12 mm)].
* Case 2: 200 x 150 mm - the control damper consists of one adjustable blade and the difference will be covered with a fixed blade (final measurements - 200 x 150 mm).

Accessories
Drive actuator

Materials and finishing
Natural aluminum material for frame and blades
Polypropylene sprocket wheels

Technical design



Assembly Methods
Screws on the mounting flange.

Selection table VCD

Air flow	Size [LxH]	800x500		1000x500		1500x500		800x800		1000x800		1200x800		1500x800		1200x1200		1500x1200		1800x1200		2000x1200		2200x1200	
(m^3/h)	α [°]	0	45	0	45	0	45	0	45	0	45	0	45	0	45	0	45	0	45	0	45	0	45	0	45
1000	Veff [m/s]	1.3	3.7																						
	△Pt [pa]	1.3	8.8																						
1500	Veff [m/s]	1.9	5.5																						
	△Pt [pa]	2.5	19.4																						
2000	Veff [m/s]	2.5	7.4																						
	△Pt [pa]	4.2	34.3																						
2500	Veff [m/s]	3.1	9.2																						
	△Pt [pa]	6.5	53.5																						
3000	Veff [m/s]	3.8	11.1	1.9	5.5																				
	△Pt [pa]	9.2	76.8	2.5	19.4																				
3500	Veff [m/s]	4.4	12.9	2.2	6.5																				
	△Pt [pa]	12.4	104.5	3.3	26.3																				
4000	Veff [m/s]	5.0	14.8	2.5	7.4																				
	△Pt [pa]	16.1	136.4	4.2	34.3																				
4500	Veff [m/s]	5.7	16.6	2.8	8.3	1.9	5.5																		
	△Pt [pa]	20.3	172.5	5.3	43.4	2.5	19.4																		
5000	Veff [m/s]	6.3	18.4	3.1	9.2	2.1	6.2																		
	△Pt [pa]	25.0	212.9	6.5	53.5	3.0	23.9																		
5500	Veff [m/s]			3.5	10.1	2.3	6.8																		
	△Pt [pa]			7.8	64.6	3.6	28.9																		
6000	Veff [m/s]			3.8	11.1	2.5	7.4																		
	△Pt [pa]			9.2	76.8	4.2	34.3																		
6500	Veff [m/s]			4.1	12.0	2.7	8.0																		
	△Pt [pa]			10.7	90.1	4.9	40.2																		
7000	Veff [m/s]			4.4	12.9	2.9	8.6																		
	△Pt [pa]			12.4	104.5	5.7	46.6																		
7500	Veff [m/s]					3.1	9.2																		
	△Pt [pa]					6.5	53.5																		
8000	Veff [m/s]					3.4	9.8																		
	△Pt [pa]					7.3	60.8																		
8500	Veff [m/s]					3.6	10.5																		
	△Pt [pa]					8.2	68.6																		
10000	Veff [m/s]							5.0	14.6	4.0	11.7	3.3	9.7	2.7	7.8	2.2	6.5								
	△Pt [pa]							15.7	132.8	1.0	85.1	0.7	59.2	4.7	38.0	3.4	26.8								
11000	Veff [m/s]							5.5	16.0	4.4	12.8	3.6	10.7	2.9	8.5	2.4	7.2								
	△Pt [pa]							18.9	160.6	12.2	102.9	8.6	71.5	5.6	45.9	4.0	32.3								
15000	Veff [m/s]									6.0	17.5	5.0	14.6	4.0	11.7	3.3	9.8	2.7	7.8	2.2	6.5	2.0	5.9		
	△Pt [pa]									22.4	191.1	15.7	132.8	10.1	85.1	7.2	59.9	4.7	38.4	3.4	26.8	2.8	21.7		
16000	Veff [m/s]									6.4	18.6	5.3	15.5	4.2	12.4	3.6	10.4	2.8	8.3	2.4	6.9	2.1	6.3	1.9	5.6
	△Pt [pa]									25.5	217.3	17.8	151.1	11.5	96.8	8.2	68.1	5.3	43.7	3.8	30.4	3.1	24.7	2.5	20.1
17000	Veff [m/s]									6.8	19.8	5.6	16.5	4.5	13.2	3.8	11.1	3.0	8.9	2.5	7.4	2.3	6.6	2.0	6.0
	△Pt [pa]									28.7	245.3	20.0	170.4	12.9	109.2	9.2	76.8	6.0	49.3	4.2	34.3	3.5	27.8	2.9	22.5
18000	Veff [m/s]											6.0	17.5	4.8	14.0	4.0	11.7	3.2	9.4	2.7	7.8	2.4	7.0	2.1	6.3
	△Pt [pa]											22.4	191.1	14.5	122.4	10.3	86.1	6.7	55.2	4.7	38.4	3.9	31.2	3.2	25.4
20000	Veff [m/s]											6.6	19.4	5.3	15.5	4.4	13.0	3.6	10.4	3.0	8.7	2.7	7.8	2.4	6.2
	△Pt [pa]											27.6	235.8	17.8	151.0	12.6	106.2	8.2	68.1	5.8	47.4	4.7	38.4	3.8	31.1
30000	Veff [m/s]															6.7	19.5	5.3	15.6	4.4	13.0	4.0	11.7	3.6	10.5
	△Pt [pa]															28.0	238.5	18.0	152.8	12.6	106.2	10.3	86.1	8.4	69.8
35000	Veff [m/s]																	6.2	18.2	5.2	15.2	4.7	13.7	4.2	12.3
	△Pt [pa]																	24.4	207.8	17.0	144.4	13.8	117.0	11.2	94.8
40000	Veff [m/s]																			5.9	17.4	5.3	15.6	4.8	14.1
	△Pt [pa]																			22.1	188.5	18.0	152.8	14.7	123.9
45000	Veff [m/s]																			6.7	19.5	6.0	17.6	5.4	15.8
	△Pt [pa]																			28.0	238.5	22.7	193.3	18.4	156.7

Annotation

* α [°] - Closing angle of the blades
* Veff[m/s] - Air velocity through the damper
* ΔPt [pa] - Pressure loss
* *Veff [m/s] = Air flow [mc/h] / 3600[s] / Ak[m^2]

Order codes

Type	/	S	/	L x H
VCD		Actuator		On request